

MAHESHCHANDRA HEGDE

Full Stack Architect & Technology Leader | 18+ Years Experience

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Technical Validation Response: Concrete, authentic responses grounded directly in 18+ years of hands-on software engineering, enterprise modernization, and product architecture. Ready for evaluation via **PramitiHR.ai**.

CANDIDATE PROFILE & LOGISTICS CONFIRMATION

Evaluation Parameter	Candidate Confirmation
Current Location	Bangalore, India
Work from Bangalore (Hybrid)	Yes (100% comfortable with working in Bangalore in a hybrid model)
Notice Period	30 Days / Immediate to 15 Days (Negotiable based on onboarding schedule)
Current & Expected CTC	Current: [Available upon request] Expected: [Market Standard for Principal Architect]
LinkedIn Profile	https://www.linkedin.com/in/maheshchandrahegde/
Updated Resume	Attached alongside this submission (CV_maheshchandra_hegde.pdf)
Hands-on vs. Architecture Split	65% Hands-on Coding 35% System Architecture & Mentorship
Product / SaaS Engineering	8-10+ Years dedicated SaaS platform engineering (18+ Years overall)

TECHNICAL VALIDATION & DOMAIN EXPERTISE

1 Hands-on Experience: TypeScript, Node.js, React.js, Next.js

- TypeScript (7+ Years):** Extensive hands-on experience using TypeScript across frontend applications (React, Next.js, Angular, Web Components) and Node.js backend services. Heavy emphasis on strict type safety, shared interfaces between client/server, and maintaining scalable component codebases.
- Node.js (10+ Years):** Extensive experience building backend REST APIs, Backend-for-Frontend (BFF) layers, asynchronous event-driven services, WebSocket servers, and developer build tooling.
- React.js (8+ Years):** Enterprise and production React development dating from early migrations at Cisco Systems (React 0.14/15) to mission-critical healthcare software at Philips (ICCA), Moonraft design systems, and modern React 18+ (Hooks, Context, Zustand, Redux Toolkit, and performance profiling).
- Next.js (4+ Years):** Architecting modern SaaS web interfaces and marketing portals utilizing Server-Side Rendering (SSR), Static Site Generation (SSG), App Router, and API routes.

2 Backend Frameworks: NestJS, Express, Fastify

- Express (8+ Years):** My primary and most extensively used Node.js web framework across production projects. Built RESTful API gateways, authentication middleware, rate limiting, and backend proxy services.
- NestJS (2+ Years):** Solid hands-on and architectural experience using NestJS for modular, enterprise-grade TypeScript backends—leveraging dependency injection (DI), modular structure, Guards for authorization, and Pipes for request validation.
- Fastify:** Conceptual and working familiarity with Fastify's schema-driven routing and low-overhead benchmarks, though Express and NestJS remain my primary production choices.

3 Data & Communication: GraphQL, gRPC, PostgreSQL, MongoDB

Across my product and enterprise platforms (including **Shutlifty Sports OS**, **eBodhya Academic OS**, and enterprise modernizations at **Cisco** and **Philips**), I tailor data and communication layers to specific workload patterns:

- PostgreSQL & MongoDB:** I leverage **PostgreSQL** as the core transactional backbone for ACID compliance, relational integrity, and structured domain state (tournament draws, institutional records), complemented by **MongoDB** (and NoSQL document stores) for high-velocity match scoring, flexible dynamic schemas, and unstructured event telemetry.

- **GraphQL & gRPC:** I implement **GraphQL** to aggregate disparate backend services and eliminate over/under-fetching across heterogeneous web dashboards and mobile/spectator interfaces, while utilizing **gRPC & Protocol Buffers** for contract-driven, low-latency binary serialization and inter-service communication in distributed microservice architectures.

END-TO-END ARCHITECTURE OWNERSHIP (HLD & LLD)

1. VoxelForge AI & Ayam3d (Generative 3D & Spatial AI Platforms — Technoyana / SrushtiLabs)

Spatial AI React / Next.js Three.js / WebGL Node.js Mesh Pipeline

- **Architecture Overview:** Architected end-to-end generative 3D platforms converting text prompts into optimized, game-ready 3D assets and parametric meshes.
- **Key Decisions:** Designed a modern web client (React/Next.js, TypeScript) coupled with asynchronous Node.js microservices and WebGL/Three.js interactive inspection viewers; engineered automated mesh optimization (retopology and polygon decimation) to deliver real-time browser performance.

2. Cisco Stadium Vision Director Modernization (UST Global / Cisco Systems)

Strangler-Fig Migration Micro-Frontends Angular & React Zero-Downtime

- **Architecture Overview:** Enterprise digital video distribution and live display management system deployed across major international sports stadiums.
- **Key Decisions:** Led the incremental modernization from legacy Adobe Flash/Flex to modular Angular and React micro-frontends. Architected an inter-runtime event bridge enabling legacy and modern components to run concurrently, ensuring zero downtime during live stadium operations.

3. Philips IntelliSpace Critical Care & Anesthesia — ICCA (Cyient / Philips Healthcare)

Healthcare 24/7 ICU Web Workers Zero Memory Leaks Strict RBAC/MFA

- **Architecture Overview:** Mission-critical clinical software suite deployed in hospital ICUs for continuous 24/7 patient telemetry and monitoring.
- **Key Decisions:** Architected the web-tier telemetry dashboard using React and TypeScript; offloaded high-density streaming calculations to Web Workers to ensure a locked 60 FPS and zero memory leaks under non-stop bedside operation, while enforcing strict hospital security (MFA, RBAC, clinical compliance).

ARCHITECTURE TRADE-OFFS & CRITICAL DESIGN DECISIONS

• Trade-off 1: Offline-First Local Storage vs. Cloud-Only WebSockets (Shutlify Sports OS)

Challenge: Relying solely on real-time cloud WebSockets failed during venue network dropouts, freezing umpire scoring tablets during live games.

Decision & Trade-off: Architected local IndexedDB write-ahead logging with optimistic UI updates, syncing asynchronously to the cloud. Accepted higher client-side sync and conflict-handling complexity to guarantee 100% court-side scoring continuity.

• Trade-off 2: Incremental Strangler-Fig Migration vs. Greenfield Full Rewrite (Cisco Stadium Vision)

Challenge: Complete rewrite proposals risked a 2-year feature freeze and high risk of regression for major stadiums.

Decision & Trade-off: Introduced hybrid micro-frontend bridges so new React/Angular modules could coexist with legacy Flash SWFs. Accepted temporary dual-runtime overhead to protect client operations and deliver incremental value every sprint with zero stadium downtime.

• Trade-off 3: Web Workers + Virtualized Canvas vs. Standard React State (Philips ICCA)

Challenge: Streaming 50Hz clinical telemetry directly into React component state caused excessive re-renders, thread locking, and memory accumulation.

Decision & Trade-off: Shifted telemetry buffering and metric calculation into Web Workers, rendering to canvas with typed arrays. Sacrificed idiomatic React state simplicity in exchange for rock-solid 60 FPS and zero memory leaks in 24/7 ICU environments.

CLAUDE, CLAUDE CODE & AI-ASSISTED ENGINEERING DEPTH

7 Experience with Claude, Claude Code, and AI-Assisted Engineering

- **Daily Engineering Workflow:** Active daily user of **Claude 3.5 Sonnet / 3.7 Sonnet**, **Claude Code CLI**, and AI coding assistants for terminal-driven workflows, multi-file refactoring, writing comprehensive unit/integration test suites (Jest/Vitest), and drafting technical design documentation.
- **Hands-on AI Products Architected:**
 - **Technoyana / SrushtiLabs (VoxelForge AI & Ayam3d):** Built generative AI platforms that turn natural language prompts into modular 3D assets, automated retopology, and tileable PBR texture workflows (srushtilabs.com/voxelforge).
 - **eBodhya Technologies:** Architected an Academic Intelligence OS incorporating LLM-assisted workflows for curriculum-aligned question generation, rubric-based evaluation, and student learning analytics.
- **AI Best Practices:** Strong practical experience with structured JSON outputs (tool calling / response schemas), prompt chaining, context window management, and hallucination guardrails.

8 Reviewing, Validating & Improving AI-Generated Architecture & Code

- **Example 1: Fixing Concurrency & Race Conditions in AI-Generated Logic:** An LLM generated a match scheduling service using naive read-then-write logic (`SELECT ... THEN UPDATE`). Identified that concurrent court updates would cause overlapping bookings and race conditions; refactored the logic to use atomic database transactions and row-level locks (`SELECT ... FOR UPDATE`).
- **Example 2: Eliminating AI Security Vulnerabilities & Deprecated Packages:** An AI model proposed an authentication and document upload pipeline using outdated npm libraries and relying on client-supplied MIME types. Replaced with server-side magic-byte inspection, secure RS256 JWT validation, and automated dependency vulnerability scanning (npm audit / Snyk).
- **Example 3: Resolving Memory Leaks in AI-Generated Frontend Loops:** An AI-generated 3D WebGL/Three.js rendering routine was recreating geometries and materials on every animation frame, leading to heavy garbage collection stutter and browser crashes. Re-engineered it with object pooling and explicit resource disposal (`dispose()`), reducing memory consumption by 80%.

DEVOPS REALISM, SAAS EXPERIENCE & GLOBAL CLIENTS

9 Practical DevOps Scope: Docker, Cloud, CI/CD, Kubernetes & Terraform

- **Honest & Authentic Scope:** My core expertise is in **Full Stack Application Architecture, Frontend Systems, Node.js Backends, and Product Engineering**, rather than dedicated Cloud Infrastructure / SRE engineering.
- **Hands-on Exposure:**
 - **Containerization & Cloud:** Regularly build multi-stage Dockerfiles for optimized Node.js and frontend production builds. Hands-on deploying and running applications on AWS (S3, CloudFront, EC2, ECS) and GCP/Firebase.
 - **CI/CD Automation:** Setting up automated continuous integration and delivery pipelines via GitHub Actions and GitLab CI.
 - **Kubernetes, Helm, ArgoCD, Terraform, Istio:** Solid architectural and conceptual understanding gained from collaborating closely with enterprise DevOps teams (at Cisco, Philips, and Ness). I review deployment manifests, configure environment variables, and ensure container health checks and ingress routing align with application requirements, while working alongside dedicated DevOps engineers for cluster/mesh management.

10 & 11 SaaS Experience & Japanese / Global Enterprise Background

- **8–10+ Years Product/SaaS Engineering:** End-to-end multi-tenant SaaS lifecycle leadership across Technoyana, Twitan (Shutliff Sports OS), SrushtiLabs (VoxelForge AI), eBodhya, Cisco Stadium Vision, and Philips ICCA (18+ Years overall engineering experience).
- **Japanese Corporate Experience (Obayashi Corporation):** Direct technical engagement with Japanese leadership at **Obayashi Corporation** (premier Japanese general contractor and engineering conglomerate). Developed comprehensive technical architecture presentations, executive work experience dossiers, and systems proposals specifically tailored to Japanese enterprise standards—emphasizing meticulous quality (*monozukuri* mindset), precision documentation, robust security, and long-term maintainability.
- **Global Enterprise Experience:** 4+ years collaborating with **Cisco Systems (USA)**, **Philips Healthcare (Netherlands/Global)**, **Vodafone (UK/Europe)**, and **CAE (Canada)**. Holder of a Master of Science (M.S.) in Computing from **Robert Gordon University, Scotland, UK**.

Closing Note: Ready to complete the **PramitiHR.ai** AI skill evaluation immediately upon link dispatch. Full architectural portfolio, GitHub repositories, and code samples available upon request.

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